

EXPERIMENT 17: SUM OF INTEGERS FROM 1 TO N

AIM

To write a Prolog program to calculate the sum of integers from 1 to N using recursion.

OBJECTIVE

- To understand recursion in Prolog.
- To calculate the sum of a sequence of integers.
- To use facts and rules for solving a mathematical problem.

ALGORITHM

1. Start.
2. Read the value of N.
3. If $N = 0$, return sum as 0.
4. Otherwise, calculate the sum of $N-1$.
5. Add N to the previous sum.
6. Display the result.
7. Stop.

SOURCE CODE

```
% Sum of integers from 1 to N
```

```
sum_to_n(0, 0).
```

```
sum_to_n(N, Sum) :-
```

```
    N > 0,
```

```
    N1 is N - 1,
```

```
    sum_to_n(N1, S1),
```

```
    Sum is N + S1.
```

Query

```
?- sum_to_n(10, Sum).
```

OUTPUT

```
?-  
% c:/Users/dloge/Desktop/pr1/1.pl compiled 0.00 sec, 2 clauses  
?- sum(5,Sum).  
Sum = 15 ,  
  
?- sum(10,Sum).  
Sum = 55 ,  
  
?-
```

RESULT

Sum = 55.

The Prolog program successfully calculates the sum of integers from **1 to N**.